Get the equation of a circle when given 3 points

Asked 11 years, 10 months ago Modified today Viewed 233k times

40

Get the equation of a circle through the points $(1, 1), (2, 4), (5, 3)$.

I can solve this by simply drawing it, but is there a way of solving it (easily) without having to draw?

analytic-geometry

Share Cite Edit Follow Flag

asked Oct 14, 2012 at 15:07
JohnPhteven
2,047 4 33 50

check out stackoverflow.com/questions/62488827/... as well – Hope Feb 9, 2022 at 7:28

@kebs you can always try the wayback machine for finding old websites: web.archive.org/web/20210503153113/https://www.qc.edu.hk/math/... – Nathan May 27, 2022 at 16:14

17 Answers

Sorted by: Highest score (default)

56

Follow these steps:

1. Consider the general equation for a circle as $(x - x_c)^2 + (y - y_c)^2 - r^2 = 0$
2. Plug in the three points to create three quadratic equations

$$(1 - x_c)^2 + (1 - y_c)^2 - r^2 = 0$$

$$(2 - x_c)^2 + (4 - y_c)^2 - r^2 = 0$$

$$(5 - x_c)^2 + (3 - y_c)^2 - r^2 = 0$$

3. Subtract the first from the second, and the first from the third to create two linear equations

$$-2x_c - 6(y_c - 3) = 0$$

$$(y_c + 7) - 6x_c = 0$$

4. Solve for the center as

$$(x_c, y_c) = (3, 2)$$

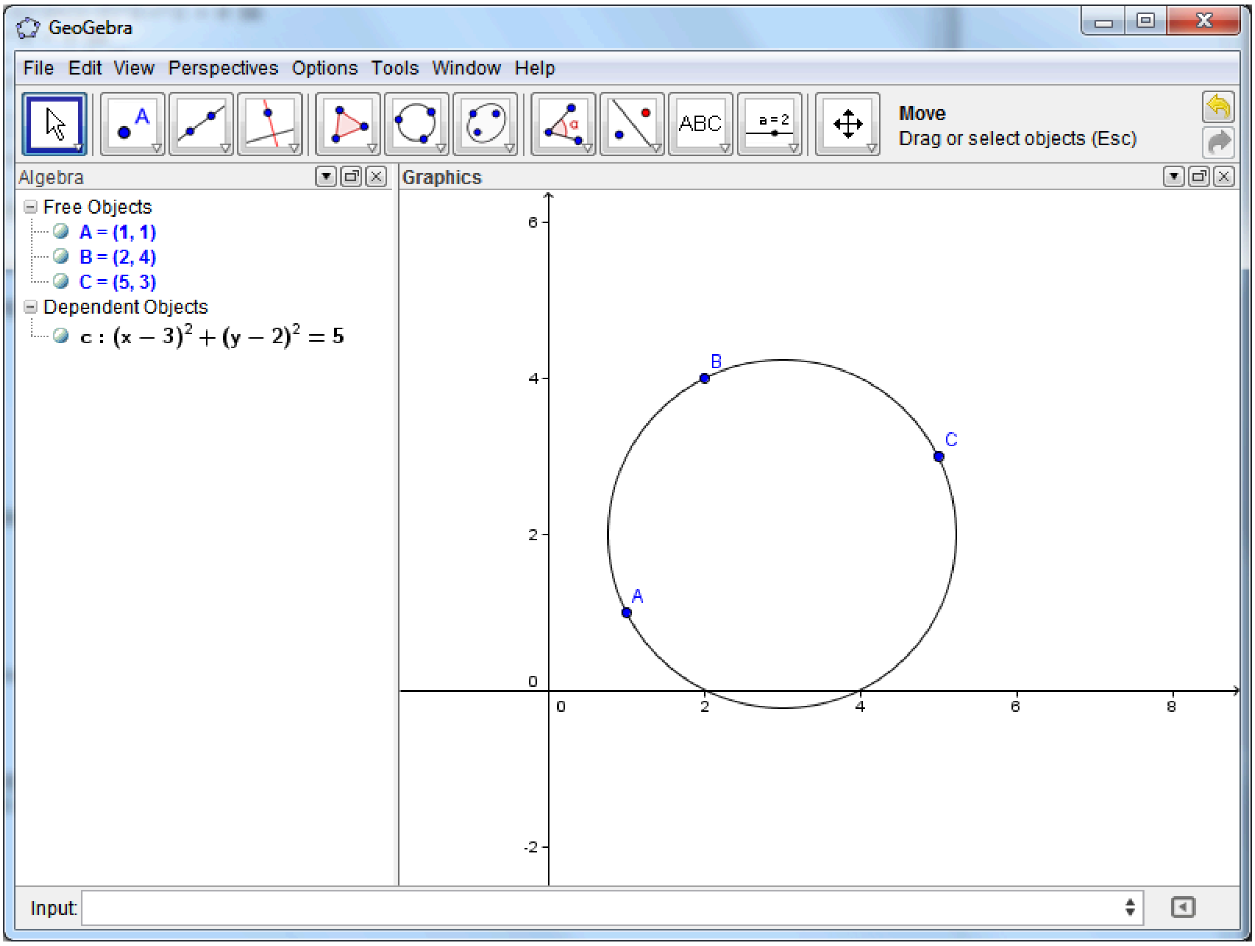
5. Plug the values for the center in any of the three quadratic equations above (I choose the first) and solve for r

$$(1 - 3)^2 + (1 - 2)^2 - r^2 = 0$$

$$5 - r^2 = 0$$

$$r = \sqrt{5}$$

6. Verify result with GeoGebra (optional)



Share Cite Edit Follow Flag

answered Oct 14, 2012 at 15:33

John Alexiou
14.2k 1 37 73

29

Based on what I've learnt from this page by Stephen R. Schmitt (thanks to @Dan_Uznanski for pointing the concept out), there's a faster way to do it using matrices:

let $\langle x_1, y_1\rangle$, $\langle x_2, y_2\rangle$, $\langle x_3, y_3\rangle$ be your 3 points, and let $\langle x_0, y_0\rangle$ represent the center of the circle.

$$\text{let } A = \begin{bmatrix} x^2+y^2 & x & y & 1 \\ x_1^2+y_1^2 & x_1 & y_1 & 1 \\ x_2^2+y_2^2 & x_2 & y_2 & 1 \\ x_3^2+y_3^2 & x_3 & y_3 & 1 \end{bmatrix} = \begin{bmatrix} x^2+y^2 & x & y & 1 \\ 1^2+1^2 & 1 & 1 & 1 \\ 2^2+4^2 & 2 & 4 & 1 \\ 5^2+3^2 & 5 & 3 & 1 \end{bmatrix} = \begin{bmatrix} x^2+y^2 & x & y & 1 \\ 2 & 1 & 1 & 1 \\ 20 & 2 & 4 & 1 \\ 34 & 5 & 3 & 1 \end{bmatrix}$$

$$\text{Note: } M_{11} = 0 \implies \langle x_0, y_0\rangle \text{ is undefined} \wedge \text{the points lie on a line rather than a circle.}$$

$$x_0 = \frac{1}{2}\cdot\frac{M_{12}}{M_{11}} = \frac{1}{2}\cdot \begin{vmatrix} 2 & 1 & 1 \\ 20 & 4 & 1 \\ 34 & 3 & 1 \end{vmatrix} \Bigg/ \begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 5 & 3 & 1 \end{vmatrix} = \frac{1}{2}\cdot\frac{-60}{-10} = 3$$

$$y_0 = -\frac{1}{2}\cdot\frac{M_{13}}{M_{11}} = -\frac{1}{2}\cdot \begin{vmatrix} 2 & 1 & 1 \\ 20 & 2 & 1 \\ 34 & 5 & 1 \end{vmatrix} \Bigg/ \begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 5 & 3 & 1 \end{vmatrix} = -\frac{1}{2}\cdot\frac{40}{-10} = 2$$

The radius, r, can then be calculated with Pythogoras' theorem, or using matrices again:

$$r^2 = x_0^2 + y_0^2 + \frac{M_{14}}{M_{11}}$$

- $M_{12}(A)$ is a minor of A — the determinant of A without row 1 or column 2.
- Lines on either side of a matrix indicate the determinant (e.g. $|A|$) (sometimes written det (A))

If you have the time, it's really worth learning about matrices — they make a lot of things much faster, and are just generally awesome! KhanAcademy has a pretty easy introduction to matrices and linear algebra.

edited Jul 8, 2022 at 6:13
zneak
721 1 5 18

answered Oct 1, 2015 at 20:04
Zaz
1,496 1 18 34

1 I cross-confirmed that the above works. However, 1 + 1 = 2.

$$\begin{bmatrix} x^2+y^2 & x & y & 1 \\ \mathbf{2} & 1 & 1 & 1 \\ 20 & 2 & 4 & 1 \\ 34 & 5 & 3 & 1 \end{bmatrix}$$

So $x_0 = 3$, $y_0 = 2$, and $r^2 = 5$. – Bob Veats Aug 9, 2016 at 17:41

3 Since the link in the post seems to be dead, here is a link to a version in the Wayback Machine. – Martin Sleziak Dec 10, 2018 at 18:55

1 Thank you @Martin and zneak for your kind contributions. The answer is now fixed. – Zaz Jul 8, 2022 at 6:14

26

Coming back to this question because I was looking for ways to do this quickly on a computer without matrix math or systems of equations. A derivation follows, with the result being only 3 lines of simple code (plus error checking).

This problem becomes quite easy if we map each pair of (x, y) points to three points in the complex plane of the form $z = x + iy$. We now apply the linear transformation $z \to \frac{z-z_1}{z_2-z_1}$, which transforms the unique points $[z_1, z_2, z_3]$ to the points $[0, 1, w = \frac{z_3-z_1}{z_2-z_1}]$.

We now square the relation $|z - c| = r$ for each transformed point to find the following set of three equations:

$$|c|^2 = r^2$$

$$1 - c - \bar{c} + |c|^2 = r^2$$

$$|w|^2 - \bar{w}c - w\bar{c} + |c|^2 = r^2$$

Subtracting the first from the second and third equations gives us the series of equations:

$$\begin{bmatrix} 1 & 1 \\ \bar{w} & w \end{bmatrix} \begin{bmatrix} c \\ \bar{c} \end{bmatrix} = \begin{bmatrix} 1 \\ |w|^2 \end{bmatrix}$$

Invert the left hand side and multiply to get:

$$\begin{bmatrix} c \\ \bar{c} \end{bmatrix} = \frac{1}{w - \bar{w}} \begin{bmatrix} w - |w|^2 \\ -\bar{w} + |w|^2 \end{bmatrix}$$

Note that this is singular if and only if $w = \bar{w}$, which is to say if w is real, which means that the points $[0, 1, w]$ are collinear on the real axis. And since we applied a linear transformation, this only happens if the original points $[z_1, z_2, z_3]$ were collinear in the complex plane to begin with!

The center of the circle can now be read off from the first row:

$$c' = \frac{w - |w|^2}{w - \bar{w}}$$

But remember that we need to undo the linear transformation, so here is the answer in our original coordinates:

$$c = (z_2 - z_1)\frac{w - |w|^2}{w - \bar{w}} + z_1$$

The radius r can be found from the equation before.

Or to make this nice and tidy, here is how you solve the problem in a few lines of python (no imports needed):

```
def circle_from_3_points(z1:complex, z2:complex, z3:complex) -> tuple[complex, float]:
    if (z1 == z2) or (z2 == z3) or (z3 == z1):
        raise ValueError(f"Duplicate points: {z1}, {z2}, {z3}")

    w = (z3 - z1)/(z2 - z1)

    # You should change 0 to a small tolerance for floating point comparisons
    if abs(w.imag) <= 0:
        raise ValueError(f"Points are collinear: {z1}, {z2}, {z3}")

    c = (z2 - z1)*(w - abs(w)**2)/(2j*w.imag) + z1  # Simplified denominator
    r = abs(z1 - c)

    return c, r


>> print(circle_from_3_points(1+1j, 2+4j, 5+3j))  # Can also generate 1+1j with `complex(1,1)`
     ((3+2j), 2.2360679775)

>> print(circle_from_3_points(1+1j, 1+1j, 3+3j))
     ValueError: Duplicate points: (1+1j), (1+1j), (3+3j)

>> print(circle_from_3_points(1+1j, 2+2j, 3+3j))
     ValueError: Points are collinear: (1+1j), (2+2j), (3+3j)
```

Thanks to David Wright of Oklahoma State University for this method.

Share Cite Edit Follow Flag

edited Feb 21, 2023 at 22:45

answered Jan 9, 2020 at 20:22

Scott
383 3 7

3 this answer should have more upvotes... – jwezorek Mar 29, 2021 at 14:45

Link gives a 404. – ddejohn Sep 17, 2021 at 21:53

Can't find that page in the wayback machine or googling around, I'll leave it there in case someone else has better luck in the future. The math is still good though :) – Scott Sep 18, 2021 at 0:26

24

I'm surprised this one hasn't been mentioned; you can find the equation by using the determinant of a matrix:

$$\begin{vmatrix} x^2+y^2 & x & y & 1 \\ 1^2+1^2 & 1 & 1 & 1 \\ 2^2+4^2 & 2 & 4 & 1 \\ 5^2+3^2 & 5 & 3 & 1 \end{vmatrix} = 0$$

This gives the equation of the circle through those three points. This sort of thing can be used in a lot of situations: matrix-determinant solutions are available for any shape I can think of where you're given points that land on the shape.

Implementing this on a computer involves having a thing that calculates determinants; to do it numerically you'll need to apply the cofactors method to avoid multiplying the variables in.

Share Cite Edit Follow Flag

answered Feb 12, 2015 at 3:54

Dan Uznanski
11.3k 2 26 42

One of my favorite tricks - extends to more general conics as well! – Mark McClure Apr 7, 2015 at 22:03

1 @Dan Uznasnski Can this method be extended to a circle defined by three points (x, y, z) in space? – nonremovable Apr 6, 2017 at 13:19

I just tried it and don't like it. I'd suggest using 3d plane operations: you can find the center by intersecting the perpendicular bisector planes of a, b and a, c: $(a-b)\cdot v = (a-b)\cdot(a+b)/2$ and the common plane that contains a, b, c. This gives you the center, and then distance to one of the points gives you a radius. The circle is then the intersection of the sphere and the common plane. – Dan Uznanski Apr 6, 2017 at 21:37

15

You can also find first R from the sin Law:

$$R = \frac{BC}{2\sin(A)} = \frac{BC \cdot AB \cdot AC}{2\|AB \times AC\|} \qquad (*)$$

Next, write the equations of circles of radius R with centre A and B and solve.

Note The formula $(*)$ is the well known geometric formula for the area of a triangle:

$$\text{Area} = \frac{abc}{4R}.$$

Share Cite Edit Follow Flag

edited Feb 12, 2015 at 3:30
Michael Hardy
1

answered Oct 14, 2012 at 15:47
N. S.
133k 12 146 267

This was just what I was looking for. This kind of formula was quick to implement. stackoverflow.com/a/26903599/999943 – phyatt Nov 13, 2014 at 7:51

Is AB denoting a vector or a length? – Christian Lindig Mar 3, 2021 at 16:56

14

As a programmer this was the best solution for me as there is no division by zero:

For given (x_1, y_1), (x_2, y_2) and (x_3, y_3) first form A, B, C and D:

$$A = x_1(y_2 - y_3) - y_1(x_2 - x_3) + x_2y_3 - x_3y_2$$

$$B = (x_1^2 + y_1^2)(y_3 - y_2) + (x_2^2 + y_2^2)(y_1 - y_3) + (x_3^2 + y_3^2)(y_2 - y_1)$$

$$C = (x_1^2 + y_1^2)(x_2 - x_3) + (x_2^2 + y_2^2)(x_3 - x_1) + (x_3^2 + y_3^2)(x_1 - x_2)$$

$$D = (x_1^2 + y_1^2)(x_3y_2 - x_2y_3) + (x_2^2 + y_2^2)(x_1y_3 - x_3y_1) + (x_3^2 + y_3^2)(x_2y_1 - x_1y_2)$$

If A=0 then the points are colinear elasewhere using A,B,C you can find center and radius of circle:

$$x_c = -B/2A$$

$$y_c = -C/2A$$

$$R = \sqrt{\frac{B^2 + C^2 - 4AD}{4A^2}}$$

Finally the formula of the circle is:

$$(x - x_c)^2 + (y - y_c)^2 = R^2$$

Share Cite Edit Follow Flag

edited Apr 19 at 15:39
Skdmg
640 1 23

answered Jan 26, 2021 at 19:16
Ali Sheikhpour
251 3 7

The right thing to say about A is that "if $A \neq 0$ then the points are not colinear". So, when $A \neq 0$ the centre and the radius are well defined, as you state. But there is a closed family of non-colinear points for which $A = 0$ and your formulation doesn't work. I though it was worth mentioning it. – MonLau Aug 30 at 10:12

10

Big hint:

Let $A \equiv (1,1)$,$B \equiv (2,4)$ and $C \equiv (5,3)$.

We know that the perpendicular bisectors of the three sides of a triangle are concurrent.Join A and B and also B and C.

The perpendicular bisector of AB must pass through the point $(\frac{1+2}{2}, \frac{1+4}{2})$

Now find the equations of the straight lines AB and BC and after that the equation of the perpendicular bisectors of AB and BC.Solve for the equations of the perpendicular bisectors of AB and BC to get the centre of your circle.

Share Cite Edit Follow Flag

answered Oct 14, 2012 at 15:19
Richard Nash
639 8 21

3 Ok, this is my answer: Line segment AB has the formula $AB = 0,5x + 0,5$, so its bisector has $y = -2x + 8$ and Line segment AC has formula $AC = 3x - 2$, so its bisector has $y = -\frac{1}{3}x + 3$. Look for the intersection between the lines: $(3, 2)$. From there it's easy to find that $r = \sqrt{5}$, so $(x - 3)^2 + (y - 2)^2 = 5$ – JohnPhteven Oct 14, 2012 at 15:26

1 You seem to be on the right track.Write up your answer and post it as an answer to get feedback! – Richard Nash Oct 14, 2012 at 15:28

1 I think what you left out in your explanation is that the perpendicular bisector of a line between any two points on a circle will pass through the *center* of that circle. – Jemenake Jun 9, 2016 at 14:30

4

“Plücker’s mu” provides another way to do this. Choose two of the points, $\mathbf{p}_1$ and $\mathbf{p}_2$, and let the equations of two circles that pass through those points be $f(x, y) = 0$ and $g(x, y) = 0$. Every linear combination of these two equations is also that of a circle (perhaps degenerate) that passes through these points. Then, an equation of the circle that also passes through the point $\mathbf{p}_0 = (x_0, y_0)$ is $f(x_0, y_0)g(x, y) - g(x_0, y_0)f(x, y) = 0$. This method applies generally to point sets that are given as null sets of functions.

Sometimes the coordinates are such that one can build the two required functions by inspection, but one can always use the “degenerate circle”—the line between two of the points—in this construction. For example, taking $\mathbf{p}_1 = (1, 1)$ and $\mathbf{p}_2 = (5, 3)$, we can use the circle centered at their midpoint:

$$(x - 3)^2 + (y - 2)^2 = 5$$

and the double line through these points:

$$(x - 2y + 1)^2 = 0.$$

The required equation is then

$$[(2 - 3)^2 + (4 - 2)^2 - 5](x - 2y + 1)^2 - (3 - 2 \cdot 4 + 1)^2[(x - 3)^2 + (y - 2)^2 - 5] = 0,$$

which simplifies to $(x - 3)^2 + (y - 2)^2 = 5$. (This could also have been computed in matrix form, but that would’ve required expanding the equation of the line. In this case it was easier to work directly with the equations.)

Share Cite Edit Follow Flag

answered Dec 3, 2017 at 0:19
amd
54k 3 36 91

Unbelievable method !! – Cathartic Encephalopathy Feb 24, 2021 at 18:17

One note, f and g can be general curves; needn't be circles. I found it confusing when I read your answer in the start @amd – Cathartic Encephalopathy Feb 24, 2021 at 18:19

3

Given 3 points $P_1(x_1, y_1)$, $P_2(x_2, y_2)$ and $P_3(x_3, y_3)$ on circle circumference whose center is (h,k). Circle equation for **each pair** of points can be expressed as:

$$(x_1 - h)^2 + (y_1 - k)^2 = (x_2 - h)^2 + (y_2 - k)^2,$$

and

$$(x_1 - h)^2 + (y_1 - k)^2 = (x_3 - h)^2 + (y_3 - k)^2,$$

By rearranging those equations, they can easily be put in the form of "**difference of squares**" in the form:

$$A^2 - B^2 = C^2 - D^2$$

we get :

$$(x_1-h)^2-(x_2-h)^2=(y_2-k)^2-(y_1-k)^2,$$

Where

$$A=(x_1-h),B=(x_2-h),C=(y_2-k),D=(y_1-k)$$

Solving

$$A^2-B^2=(A-B)(A+B)$$

we get:

$$[(x_1-h)-(x_2-h)][(x_1-h)+(x_2-h)]=[(y_2-k)-(y_1-k)]*[(y_2-k)+(y_1-k)]$$

simplifying:

$$(x_1-x_2)(x_1+x_2-2h)=(y_2-y_1)(y_2+y_1-2k)$$

Introducing

$$a_1=(x_1-x_2),a_1'=(x_1+x_2),b_1=(y_2-y_1),b_1'=(y_2+y_1)$$

And

$$c_1=\frac{(b_1b_1'-a_1a_1')}{2a_1}$$

we get for each pair of points :

$$h=\frac{b_1}{a_1}k-c_1$$

and another reduced equation for the other pair of points would give:

$$h=\frac{b_2}{a_2}k-c_2$$

Solving above equations , we get :

$$k=\frac{a_1a_2(c_1-c_2)}{(b_1a_2-b_2a_1)}$$

and

$$r=\sqrt{(E^2+D^2)}$$

Where

$$E=(x1-h)$$

and

$$D=(y1-k)$$

Share Cite Edit Follow Flag

edited Apr 19 at 15:39
Skdmg
640 1 23

answered Apr 10, 2019 at 11:06
Mohammad Kanan
131 4

2

Consider the general (implicit) equation that defines a circle, with parameters α, β, γ. Substitute the coordinate of the given points and get three linear equations in the three variables α, β, γ. Solve the system.

Share Cite Edit Follow Flag

answered Oct 14, 2012 at 15:12
Andrea Orta
1,132 8 18

I just wanted to flag Scott's answer to this question which is very concise and is probably faster than the matrix-based version. I've included below a C++17 implementation of the complex arithmetic solution to the problem

2

```
#include <complex>
#include <optional>
#include <iostream>

struct point {
    double x;
    double y;
};

struct circle {
    double x;
    double y;
    double r;
};

std::optional<circle> circle_through_three_points(const point& pt1, const point& pt2, const point& pt3) {

    using namespace std::complex_literals;
    auto pt_to_z = [](const point& pt)->std::complex<double> { return { pt.x,pt.y }; };
    auto z1 = pt_to_z(pt1), z2 = pt_to_z(pt2), z3 = pt_to_z(pt3);
    auto w = (z3 - z1) / (z2 - z1);

    if (w.imag() == 0)
        return std::nullopt; // the three points are collinear

    auto magnitude_w = std::abs(w);
    auto c = (z2 - z1) * (w - magnitude_w * magnitude_w) / (2i * w.imag()) + z1;
    auto r = std::abs(z1 - c);

    return circle{ c.real(), c.imag(), r };
}

int main()
{
    auto c1 = circle_through_three_points({ 1,1 }, { 2,4 }, { 5,3 });
    auto c2 = circle_through_three_points({ 1,1 }, { 2,2 }, { 3,3 });

    if (c1)
```

Share Cite Edit Follow Flag

edited Apr 20, 2021 at 1:41

answered Mar 29, 2021 at 16:23

jwezorek
267 2 9

1 Great! Don't forget to check that z1 and z2 are not equivalent, or you'll run into a divide by zero error when calculating w. – Scott Oct 4, 2021 at 22:51

This solves your problem in python...

2

```
def three_point_circle(z1,z2,z3):
    a = 1j*(z1-z2)
    b = 1j*(z3-z2)
    m1 = a.imag/a.real
    c = (z1-z2)/2
    p1 = z2+c
    b1 = p1.imag-m1*p1.real
    m2 = b.imag/b.real
    d = (z3-z2)/2
    p2 = z2+d
    b2 = p2.imag-m2*p2.real
    x = (b2-b1)/(m1-m2)
    y = (m2*b1-m1*b2)/(m2-m1)
    center = x+1j*y
    radius = abs(center-z1)
    return x,y,radius
```

as long as there is not division by zero.

If there is division by zero, you can solve this by revolving your input numbers (multiply by some random power of the imaginary unit 1j) and then revolve the x,y coordinates back (set to a complex number and multiply by 1j to the negative power you used).

Share Cite Edit Follow Flag

answered Jul 26, 2021 at 3:47

peawormsworth
289 1 7

1

(Rephrasing Scott's answer.) Let the three points be z_1, z_2, z_3 as complex numbers. These lie on a circle C_{old} given by $|z-c|^2 = r^2$. We can map C_{old} to another circle C_{new} given by $|w-d|^2 = s^2$ that is easier to work with using a linear transformation $f(z) = w = az + b$. In other words we'll have $f(C_{old}) = C_{new}$ and moreover $f(c) = d$, the center of C_{old} gets mapped by f to the center of C_{new}. This should be intuitively clear since the linear transformation translates/scales/rotates C_{old}.

One choice of linear map is $f(z) = \frac{z-z_1}{z_2-z_1}$, with $f(z_1) = w_1 = 0$, $f(z_2) = w_2 = 1$. Evaluating the defining equation for C_{new} at $w_1 = 0$, $w_2 = 1$, $w_3 = f(z_3)$ gives

$$|d|^2 = s^2 = d\bar{d}\,,$$
$$|1-d|^2 = s^2 = 1 - d - \bar{d} + |d|^2,$$
$$|w_3 - d|^2 = s^2 = |w_3|^2 - w_3\bar{d} - \bar{w}_3 d + |d|^2.$$

Simplifying some gives the system of linear equations (in $d, \bar{d}$)

$$d + \bar{d} = 1$$
$$\bar{w}_3 d + w_3 \bar{d} = |w_3|^2.$$

Solving for d gives the center of C_{new} (assuming $w_3 \neq \bar{w}_3$)

$$d = \frac{w_3 - |w_3|^2}{w_3 - \bar{w}_3},$$

and applying $f^{-1}(w) = (z_2 - z_1)w + z_1$ gives the center of C_{old}

$$c = f^{-1}(d) = z_1 + (z_2 - z_1)d.$$

[Remark: If $w_3 = \bar{w}_3$, then $z_3 - z_1 = w_3(z_2 - z_1)$ for the real scalar w_3, so that z_3 is on the line through z_1 and z_2.]

Finally, the radius of C_{old} can be found by evaluating $|z_i - c|^2 = r^2$ at any of the initial three points.

Here is yet another implementation (Python):

```
def center_radius(z1, z2, z3):
   """
   Return (center, radius), with complex center, of the circle through
   complex z1, z2, z3. If z1, z2, z3 are colinear, print "colinear" and
   return None.
   """
   if z1 == z2:
      print("colinear")
      return None

   def f(z):
      return (z-z1)/(z2-z1)

   def f_inv(w):
      return z1+(z2-z1)*w

   w3 = f(z3)
   if w3.imag == 0: #may want abs(w3.imag) < eps
      print("colinear")
      return None

   d = (w3-w3*w3.conjugate())/(w3-w3.conjugate())
   c = f_inv(d)
   r = abs(z1-c)
   return (c, r)
```

For example,

```
>>> center_radius(0,1,complex(0,1))
```

```
((0.5+0.5j), 0.7071067811865476)
```

Share Cite Edit Follow Flag

edited Aug 4, 2021 at 16:25

answered Aug 4, 2021 at 14:47

yoyo
10.1k 30 39

1

It must be unlucky to add a 13th answer to an 8-year-old question that has been viewed 131,000 times (I don't know why it appeared on the front page today), but I just have to point out that this particular example can be solved by a simple mental calculation.

If $A = (1, 1)$, $B = (2, 4)$ and $C = (5, 3)$, then - making B the "origin", to see if it helps (I tried A first, and would have tried C next, if B hadn't worked) - we have $A - B = (-1, -3)$ and $C - B = (3, -1)$, whence $(A - B) \cdot (C - B) = -3 + 3 = 0$, i.e., the angle at B is a right angle, therefore AC is a diameter, therefore the centre is $(A + C)/2 = (3, 2)$, and a radius vector is $(2, 1)$, therefore the square of the radius is 5, therefore the equation is $(x - 3)^2 + (y - 2)^2 = 25$.

Presumably the values of A, B, C were chosen to enable this simple solution.

I'll get me coat. :)

Share Cite Edit Follow Flag

answered Jul 7, 2021 at 22:13

Calum Gilhooley
12.3k 21 47

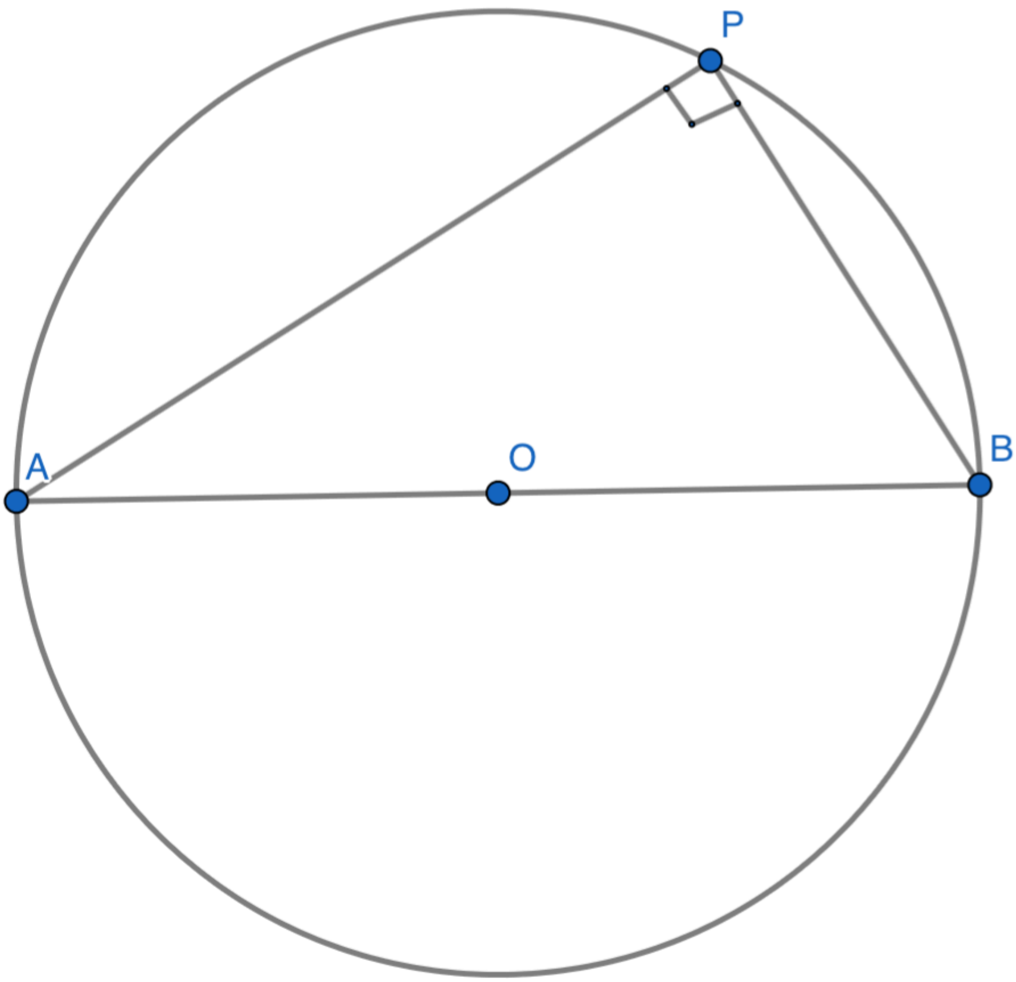
Note to self: never look at Maths.SE just before bedtime. :) – Calum Gilhooley Jul 7, 2021 at 22:19

Why the downvote? – Calum Gilhooley Dec 29, 2023 at 18:04

1

Just for the sake of adding an answer:

Take *any* two points of the given three :$A(x_1, y_1)$ and $B(x_2, y_2)$. We will first determine the circle Ω having AB as the diameter. This may seem arbitrary, but I assure you it is not.



Let $P(x, y)$ be any point on the circle. Then, $\angle APB$ is an angle in a semicircle and is hence a right angle. By writing $m_{AP} \cdot m_{BP} = -1$, we have

$$\left(\frac{y-y_1}{x-x_1}\right)\left(\frac{y-y_2}{x-x_2}\right) = -1$$

$$\implies (y-y_1)(y-y_2)+(x-x_1)(x-x_2)=0$$

is the equation of Ω. Now, the *straight line*, L, through A and B is

$$\frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1}$$

$$\implies (y-y_1)(x_2-x_1)-(x-x_1)(y_2-y_1)=0$$

Now we will write the equation of a family of circles passing through A and B. This can be written as

$$\Omega + \lambda L = 0$$

and this can be verified to being a circle for all λ. Now you can understand why we found the equation of Ω: to find a similar (to the required curve) but special curve passing through the points of interest.

Now, put (x_3, y_3) in the equation and solve for λ. This will give the required equation of the circle.

Like in the solution by amd, you use a "pencil of conics". amd is with a linear combination of 2 conics. Whereas, in your proposal, the second conic isn't a conic but a line, But in such a case shouldn't the pencil be $\Omega + \lambda L^2$ in order to have two second degree expressions (a doubled line is considered as a - degenerate - conic) ? – Jean Marie Feb 14 at 2:51

1

I know this question is outdated, but I like to show this easy solution.

First subtract the coordinates of a point from the others'. Now we are looking for the equation of a circle through the origin and two other points, which is of the form

$$(x-x_c)^2+(y-y_c)^2-r^2=x^2-2xx_c+y^2-2yy_c=0,$$

as there is no independent term.

This results in the 2×2 linear system of equations

(typo in the right hand side of the first equation of the 2x2 system changed wrong y2^2 to correct y1^2)

$$\begin{cases} 2x_cx_1+2y_cy_1=x_1^2+y_1^2, \\ 2x_cx_2+2y_cy_2=x_2^2+y_2^2. \end{cases}$$

giving the coordinates of the center and the radius $\sqrt{x_c^2+y_c^2}$. Don't forget to translate back by adding the first point.

Share Cite Edit Follow Flag

edited Aug 13, 2021 at 20:58
Robert Nowak
113 3

answered Oct 2, 2017 at 16:08
user65203

0

Equation of a circle with center (x_0, y_0) and radius r is

$$(x-x_0)^2+(y-y_0)^2=r^2$$

which can be expressed as

$$x^2+y^2+ax+by+c=0 \qquad \ldots(1)$$

where

$$\begin{aligned} x_0 &= -\tfrac{a}{2} \\ y_0 &= -\tfrac{b}{2} \\ r &= \sqrt{x_0^2+y_0^2-c} \qquad \ldots(2) \end{aligned}$$

Now, if the circle passes through points (x_1, y_1), (x_2, y_2) and (x_3, y_3), then the equation (1) is satisfied by the points, so that we have

$$\begin{aligned} ax_1+by_1+c &= -x_1^2-y_1^2 \\ ax_2+by_2+c &= -x_2^2-y_2^2 \\ ax_3+by_3+c &= -x_3^2-y_3^2 \end{aligned}$$

with solution

$$a = \frac{\begin{bmatrix} -x_1^2-y_1^2 & y_1 & 1 \\ -x_2^2-y_2^2 & y_2 & 1 \\ -x_3^2-y_3^2 & y_3 & 1 \end{bmatrix}}{\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}}$$

$$b = \frac{\begin{bmatrix} x_1 & -x_1^2-y_1^2 & 1 \\ x_2 & -x_2^2-y_2^2 & 1 \\ x_3 & -x_3^2-y_3^2 & 1 \end{bmatrix}}{\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}}$$

$$c = \frac{\begin{bmatrix} x_1 & y_1 & -x_1^2-y_1^2 \\ x_2 & y_2 & -x_2^2-y_2^2 \\ x_3 & y_3 & -x_3^2-y_3^2 \end{bmatrix}}{\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}}$$

Now, from (2), we can compute the center (x_0, y_0) and radius r of the circle.

We can implement the above with the following `r` function `find.circle()`, which accepts the arguments (x_1, y_1), (x_2, y_2) and (x_3, y_3) corresponding to the coordinates of the 3 points the circle passes through, and returns the center (x_0, y_0) and radius r of the circle:

```
find.circle <- function(x1, y1, x2, y2, x3, y3) {
  D <- matrix(c(x1, x2, x3, y1, y2, y3, 1, 1, 1), ncol=3)
  D_a <- matrix(c(-x1^2-y1^2, -x2^2-y2^2, -x3^2-y3^2, y1, y2, y3, 1, 1, 1), ncol=3)
  D_b <- matrix(c(x1, x2, x3, -x1^2-y1^2, -x2^2-y2^2, -x3^2-y3^2, 1, 1, 1), ncol=3)
  D_c <- matrix(c(x1, x2, x3, y1, y2, y3, -x1^2-y1^2, -x2^2-y2^2, -x3^2-y3^2), ncol=3)
  a <- det(D_a) / det(D)
  b <- det(D_b) / det(D)
  c <- det(D_c) / det(D)
  print(c(a,b,c))
  x0 <- -a/2
  y0 <- -b/2
  r <- sqrt(x0^2 + y0^2 - c)
  return(c(x0,y0,r))
}

find.circle(1, 1, 2, 4, 5, 3)
# [1] -6 -4  8 # (a, b, c)
# [1] 3.000000 2.000000 2.236068 # (x0, y0, r)
```

The equation of the circle is $x^2 + y^2 - 6x - 4y + 8 = 0$, i.e., $(x-3)^2 + (y-2)^2 = 5$.

Share Cite Edit Delete Flag

answered 3 mins ago

Sandipan Dey
2,131 10 12